

```
public class UniquePairs {  
    public static void printPairs(int[] arr) {  
        int n = arr.length;  
        // Outer loop tracks the first element  
        for (int i = 0; i < n; i++) {  
            // Inner loop starts from the next element to avoid duplicates  
            for (int j = i + 1; j < n; j++) {  
                System.out.println("(" + arr[i] + ", " + arr[j] + ")");  
            }  
        }  
    }  
  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3, 4};  
        printPairs(arr);  
    }  
}
```